Project Proposal: NYC Ride-Hailing, Weather & EV Charging Access

Course: DL3004 Data Warehousing & BI Lab, Fall 2026, FAST-NU Lahore **Course Instructor:** Mr. Muhammad Ishaq Raza | **Lab Instructor:** Mr. Shahzeb Mubashar **Team:** [Student Name 1 – Roll No], [Student Name 2 – Roll No], [Student Name 3 – Roll No] **Phase:** 1 – Business Analysis & Dimensional Design

1. Project Overview

We will build a data warehouse and BI pipeline using **only real public data** about ride-hailing in New York City. It joins trips, hourly weather and EV charging stations. It will show how rain changes demand, waiting time and driver pay, and which busy zones have few chargers.

2. Business Problem

Ride-hailing fleets in New York are moving to electric vehicles. Planners need to know where demand is high and whether chargers are nearby. Rain also changes demand and driver earnings. Today these facts sit in separate datasets.

Our goal: one warehouse where trips, weather and charging stations can be analysed together.

3. Objectives

- Design a star schema with 3 fact tables, using the grain-first approach from Lab 05.
- Build an ETL pipeline: Bronze (raw) → Silver (clean) → Data Warehouse → Gold (marts).
- Use SCD Type 1 and Type 2 where history matters.
- Answer business questions with OLAP SQL (ROLLUP, CUBE, window functions) and a Power BI dashboard.

4. Business Processes

Process	Fact Table	Type
Ride-hailing trips	Fact_Trips	Transaction
Hourly demand and weather per zone	Fact_Zone_Hourly_Snapshot	Periodic snapshot
Charging station openings	Fact_Charger_Coverage	Factless

Each process has its own fact table. We do not mix them.

5. Data Sources

Type	Source	What we use
Relational DB	NYC TLC High Volume FHV (Uber, Lyft) trips, with its data dictionary	500k-row sample in PostgreSQL: pickup and drop-off zone, request and pickup time, trip miles, trip time, fare, tips, driver pay
REST API	Open-Meteo	Hourly historical precipitation for NYC
JSON (API)	AFDC / NREL Alternative Fuel Stations API	EV charging stations in NYC: location, network, ports, status, open date (free API key)
Reference CSV	Taxi Zone Lookup and zone polygons	Zone names, boroughs, and shapes used to place each station in a zone

6. Planned Dimensional Model

Grain Statements

- **Fact_Trips:** One row represents one completed ride-hailing trip in the sampled data.
- **Fact_Zone_Hourly_Snapshot:** One row represents one pickup zone in one hour.
- **Fact_Charger_Coverage:** One row represents one charging station opened in one zone on one date.

Dimensions

Dimension	Key Attributes	Natural Key	SCD
Dim_Zone	Zone, borough, service zone	LocationID	**Type 1** (names corrected in place)
Dim_Station	Network, number of ports, charger level, status	station_id	**Type 2** (network, ports or status can change between data downloads)
Dim_Company	Company (Uber, Lyft, Via, Juno), base number	hvfhs_license_num	Type 1
Dim_Date	Day, month, quarter, year, weekday	date	–
Dim_TimeOfDay	Hour, peak / off-peak, day part	hour	–
Dim_Weather	Rain level (none, light, heavy)	weather_band	–

All dimensions get surrogate keys (for example `zone_key`, `station_key`). Source IDs are kept inside the dimensions. For the Type 2 history, we load an older AFDC station snapshot first and the latest data after it, so changes create new row versions.

Measures

- **Fact_Trips:** trip miles, trip time, base fare, tips, driver pay, wait time (pickup time minus request time). `trip_id` (generated row number) is a degenerate dimension.
- **Fact_Zone_Hourly_Snapshot:** trip count, total wait minutes, total driver pay, rainfall (mm). We store totals so averages are calculated as total ÷ trips, not by averaging averages.
- **Fact_Charger_Coverage:** no measures. It only records that a station opened in a zone on a date.

Hierarchies: Year → Quarter → Month → Date, and Borough → Zone.

7. Pipeline Architecture

Following the teacher's reference diagram:

1. **Sources:** PostgreSQL trips, Open-Meteo API, AFDC station API (JSON), taxi zone CSV.
2. **Bronze:** raw data loaded as it is, with batch ID and load time.
3. **Data quality checks:** NULL, data type, duplicate and referential checks. Bad rows go to a reject table with a reason.
4. **Silver:** cleaned and standardised data, loaded incrementally. Each station is placed in a zone with a one-time spatial join (Python / GeoPandas).
5. **SCD processing:** detect changes, then apply Type 1 or Type 2.
6. **Data Warehouse:** dimensions and facts, with a late-arriving data plan (unknown member rows).
7. **Gold:** data marts for trips, zone demand and charging coverage.
8. **Analytics and BI:** OLAP SQL and a Power BI dashboard with spatial maps.
9. **Audit and lineage:** run logs, row counts, and source vs warehouse reconciliation.

8. Sample Business Questions

The full list of 10 will be in our separate Business Questions PDF.

1. Which pickup zones have the most trips and the highest driver pay per mile?
2. How much does rain change trips per hour and customer wait time?
3. Which hours of the day have the longest waits?
4. Which busy zones have no charging stations?
5. How has the number of charging stations grown by borough over the years?

9. Known Challenges and Our Plan

Challenge	Our Plan
TLC trip data is very large (millions of rows)	Use a 500k-row sample
Zone matching can cause Cartesian joins	Join through unique surrogate keys on Dim_Zone
Stations only have latitude and longitude, not zones	Do a one-time spatial join with the zone polygons, then store `zone_key`

Challenge	Our Plan
Trip data does not say if a vehicle is electric	Study ride-hailing as a whole and link it to charger access by zone. We do not claim these are EV trips
Trip data has no surge multiplier	Use driver pay per mile as a simple price signal

10. Tools

PostgreSQL, pgAdmin, Python/Pandas (with GeoPandas for the zone join), Power BI Desktop, Draw.io / dbdiagram.io, GitHub.

11. Timeline

Phase	Weeks	Work	Weight
1: Design	1–3	Proposal, requirements, grain, schema, data dictionary	30%
2: Implementation	4–6	ETL, SCD, dimensions and facts	40%
3: Analytics & BI	7–8	OLAP SQL, Power BI dashboard	30%

12. Expected Outcome

A working warehouse and dashboard showing three clear findings: how rain changes demand, wait time and driver pay; which zones and hours are busiest; and where busy zones have too few chargers.